

Understanding Analysis, 2nd Edition

Chapter 5 - Section 4.6 - Sets of Discontinuity

1. Define $D_f \subset \mathbf{R}$ to be the set of points where the function f is discontinuous. Using modifications of Dirichlet / Thomae functions, construct a function $f : \mathbf{R} \rightarrow \mathbf{R}$ so that
 - (a) $D_f = \mathbf{Z}^c$.
 - (b) $D_f = \{x : 0 < x \leq 1\}$.

Solution:

- (a) Let $D(x)$ be the distance from x to the integer closest to it. That is, $D(x) = \min(|x - \lfloor x \rfloor|, |x - \lceil x \rceil|)$.

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is an integer or irrational} \\ D(x) & \text{if } x \text{ is rational} \end{cases}$$

It's continuous at integer x . If $(x_n) \rightarrow x$ then $f(x_n) \rightarrow 0$ for both rationals and irrationals in (x_n) .

It's discontinuous at rational x . Take a sequence of irrationals $(x_n) \rightarrow x$. We have $f(x_n) \rightarrow 0$ but $f(x) > 0$.

It's discontinuous at irrational x . Take a sequence of rationals $(x_n) \rightarrow x$. We have $f(x_n) \rightarrow D(x) > 0$ but $f(x) = 0$.